

Implementation of an Application for Overlapped Speech Separation

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Abstract — This technical and scientific report presents a comprehensive analysis of a complex audio signal processing system designed for vocal source separation, background noise reduction, and perceptual speech quality evaluation. The architecture of the implemented platform achieves an innovative fusion between classic Digital Signal Processing methods, such as Wiener spectral filtering, and state-of-the-art artificial intelligence models, including attention-based architecture like SepFormer and Voice Activity Detection models. Furthermore, the document investigates the evolutionary trajectory of the field, from Dual-Path networks based on segmentation to the new asymmetric architectures such as SepReformer, a recent approach that minimizes computational complexity through the early division of speaker features. The quality evaluation of the processed signals is performed through a suite of evaluation metrics, including Scale-Invariant Signal-to-Distortion Ratio, Perceptual Evaluation of Speech Quality, Extended Short-Time Objective Intelligibility, Speaker Similarity, and the non-intrusive neural metric, Deep Noise Suppression Mean Opinion Score. The document details the theoretical foundation of these methods, the software engineering logic of the application, the resource management strategies, and the three-stage comparative optimization pipeline that governs the system's operation.

Keywords — *attention mechanisms, audio separation, DNSMOS, neural networks, perceptual evaluation, SepFormer, SepReformer, signal processing, SpeechBrain.*

I. INTRODUCTION TO ACOUSTIC SOURCE SEPARATION

The problem of audio source separation, widely known in the literature as the "cocktail party" problem, is considered a major challenge in acoustic signal processing and pattern recognition [1]. Initially formulated in the 1950s by researcher Colin Cherry, this paradigm describes the human ability to focus exclusively on a single vocal source (one speaker) within a complex mixture of multiple sounds [2]. In a natural acoustic environment, a person is simultaneously subjected to competing speakers, stationary background noise (e.g., HVAC system hums), non-stationary noise (traffic, impact sounds), and reverberations caused by sound wave reflections on room surfaces. Replicating this selective filtering capability in artificial systems, using a single capture source (monaural microphone), has required decades of extensive research [3].

Blind Source Separation (BSS) and Speech Enhancement (SE) approaches have relied almost exclusively on DSP techniques applied in the time-frequency domain [3].

Established methods, such as Wiener filtering, spectral subtraction, or Non-negative Matrix Factorization (NMF), involved transforming the time-domain signal into the frequency domain using the Short-Time Fourier Transform (STFT) [4]. Although these methods are computationally extremely fast and prove effective in suppressing stationary background noise, they present significant limitations in complex environments. In scenarios with multiple overlapping speakers or strong non-stationary noise, classic DSP techniques often generate unwanted audio artifacts, known as "musical noise," and distort the essential speech components of the target voice.

The emergence and development of Deep Neural Networks (DNN) marked a significant advancement, changing audio processing methods [1]. This evolution began with the implementation of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), which demonstrated a superior capacity to map the complex characteristics of the vocal signal directly from training data [6]. Early deep learning systems still operated on time-frequency representations, where the network was trained to estimate an ideal mask (such as the Ideal Binary Mask or Ideal Ratio Mask) that was subsequently multiplied by the original spectrum. However, this reliance on the STFT introduced an important limitation: the phase reconstruction problem [7]. Separating the spectral magnitude left the phase unaltered, and using the noisy phase of the original mixed signal to reconstruct the isolated signal in the time domain invariably led to a performance limitation.

The response to this limitation came in the form of Time-Domain processing architectures, which completely eliminated the need for the STFT [8]. The evolution continued rapidly, culminating in the adaptation of self-attention mechanisms (Transformers) from natural language processing to acoustic time series [9]. This adaptation allowed the modeling of long-distance dependencies in a way impossible to achieve by classic RNNs, giving rise to reference models such as SepFormer (Attention is All You Need in Speech Separation) [9]. However, significant challenges remain, particularly regarding the high computational complexity induced by high-fidelity audio sequences, leading researchers to explore asymmetric solutions with linear complexity, such as the recent SepReformer model presented at the NeurIPS 2024 conference.

In this current research context, the software project documented in this paper represents an integrated platform, built on the PyTorch framework and the Gradio web interface, which combines classic methods with modern neural approaches. The platform does not offer just a simple access point to the models developed under the SpeechBrain aegis; it integrates a sophisticated workflow, termed the "Comparative Optimization". This hybrid pipeline integrates VAD via the

Silero model for temporal isolation, aggressive DSP filtering for cleaning the stationary noise spectrum, and finally, neural isolation based on the Transformer architecture to separate the vocal sources. Moreover, the platform implements rigorous evaluation procedures capable of objectively validating performance through intrinsic and extrinsic perceptual metrics, integrating the Microsoft DNSMOS model to emulate the decisions of a human jury [12]. The purpose of this extended report is to analyze in detail both the theoretical foundations of the involved separation architectures, the software engineering logic, the signal processing algorithms implemented in the source code, and the quantitative evaluation methodologies of auditory quality.

II. THEORETICAL FOUNDATIONS AND THE TRAJECTORY OF SEPARATION ARCHITECTURES

To fully understand the complexity and performance of the neural modules integrated into the developed platform, a detailed analysis of the evolutionary trajectory of speech separation networks is imperative. The transition from spectral mask-based approaches to fully attention-based networks represents a series of ingenious solutions to fundamental mathematical bottlenecks.

▪ Time-Domain Models and the Conv-TasNet Architecture

The first hybrid deep learning systems for voice separation processed the magnitude spectrogram of the mixed signal [7]. The standard algorithm involved calculating the STFT for the input signal, passing the magnitudes through a neural network to estimate a filtering mask, multiplying the mask with the original magnitude, and then reconstructing the waveform by applying the inverse STFT (iSTFT) using the phase of the original mixed signal [13]. Because the phase contains critical temporal information regarding the localization and structure of the waveform, using the noisy phase of the original mixture established an intrinsic upper bound on the quality of the extracted signal, a limit that could not be surpassed regardless of the neural network's size.

To avoid the phase reconstruction problem, researchers proposed moving the entire processing pipeline to the time domain. The Time-Domain Audio Separation Network (TasNet) represented a major conceptual leap [8]. TasNet completely replaced the Fourier transform with a 1D convolutional encoder (a set of learned filters) and the iSTFT with a transposed 1D convolutional decoder [8]. This linear encoder directly transforms short waveform segments (frames of a few milliseconds) into an intermediate, pseudo-spectral latent representation, which is optimized exclusively through the backpropagation process for the separation task [14].

Although TasNet demonstrated the possibility of exceeding the performance of time-frequency methods, the use of Long Short-Term Memory (LSTM) in the central separation module proved problematic. Because the convolutional encoder works optimally with very small kernel sizes (short temporal windows) to maintain minimal latency, the resulting sequence of characteristics becomes exponentially longer compared to a standard STFT spectrogram [8]. Sequences reaching 10^4 to 10^5 temporal frames overwhelmed the capacity of LSTM networks to model long-term dependencies due to the vanishing gradient and memory limitations.

This challenge led to the development of the Conv-TasNet architecture, which completely replaced the recurrent LSTM blocks with a Temporal Convolutional Network (TCN) [13]. TCNs use 1D dilated convolutions, allowing the network's receptive field to grow exponentially with the depth of the layers [13]. Conv-TasNet quickly became the industry standard, demonstrating that a fully convolutional end-to-end architecture in the time domain can far exceed ideal time-frequency masks, achieving substantial latency reductions essential for real-time applications.

▪ The Dual-Path Paradigm and Long Sequence Modeling

Despite the strong performance of Conv-TasNet, the exclusive reliance on dilated convolutions showed limitations in capturing truly global dependencies across the entire speech. The receptive window of a convolution, however dilated, remains finite, whereas speaker characteristics (such as timbre or fundamental pitch) are coherent throughout the entire audio recording [8].

To expand the modeling horizon without compromising efficiency, the Dual-Path processing paradigm was introduced, pioneered by the Dual-Path Recurrent Neural Network (DPRNN) architecture [8]. The core concept of the Dual-Path approach is a segmentation technique meant to restructure a massive one-dimensional tensor into a manageable three-dimensional matrix.

The mechanism operates as follows: the long latent sequence generated by the encoder is cut into fixed-length chunks (e.g., 250 frames) [8]. To ensure consistency at the chunk boundaries, they are typically overlapped by 50%. Then, processing unfolds iteratively on two orthogonal axes:

1. **Intra-Chunk Processing (Local):** A network layer evaluates the sequence along the temporal axis inside each individual chunk. This stage models short-duration phonetic features and local acoustic details [15].

2. **Inter-Chunk Processing (Global):** The output of the previous stage is transposed, and another network layer evaluates the sequence along the chunk axis, comparing the same temporal index across all chunks. This stage integrates the long-term global context, identifying speaker-specific characteristics throughout the recording [15].

By alternating these two paths, the DPRNN managed to efficiently model theoretically infinite signal sequences, achieving a performance that clearly outperformed Conv-TasNet. However, the use of RNNs (even on shorter segments) meant that processing remained inherently sequential, which blocked the possibility of parallelization on modern hardware architectures (GPU) and severely limited inference and training speed [10].

▪ The Transformer Revolution and the SepFormer Architecture

The solution to the parallelization limitations of RNNs came through the incorporation of the Multi-Head Self-Attention mechanism, the core of the Transformer architecture, which had already revolutionized the fields of machine translation and text processing [9]. The SepFormer model, introduced in 2021 and strongly supported by the open-source SpeechBrain library, represents a milestone in the field, being the primary model adopted in the software system documented in this report [10].

SepFormer adopts the Dual-Path conceptual framework introduced by the DPRNN, but completely replaces all recurrent blocks with Transformer modules. This modification is not trivial; it allows the network to instantaneously access any temporal frame within the chunk (via intra-chunk attention) and any chunk in the sequence (via inter-chunk attention) through a single parallelizable matrix operation [10].

The results achieved by SepFormer significantly improved previous performance levels. Based on tests performed on the WSJ0-2Mix corpus (the most widely used dataset for benchmarking speech separation, consisting of recordings from the Wall Street Journal mixed at specific signal-to-noise ratios [16]), SepFormer achieved a Scale-Invariant Signal-to-Noise Ratio improvement (SI-SNRi) of 22.3 dB, matching or surpassing all competing networks at that time [10]. Furthermore, researchers demonstrated that the inherent parallelism of Transformers allows the SepFormer model to maintain competitive performance even when the encoded representation is downsampled by a massive factor of 8, significantly reducing memory requirements and making it much faster than recurrence-based networks that achieve similar performance [10].

The integration of the SepFormer model within the analyzed software platform, achieved through the SpeechBrain library, ensures an acoustic extraction quality that meets current academic and industrial requirements. The architecture's versatility is demonstrated by the existence of multiple hyperparameter configurations within the application (such as the sepformer-whamr model for reverberant and noisy environments, or sepformer-libri3mix for three simultaneous speakers), demonstrating a high degree of generalization of the dual-path attention mechanism across various acoustic degradation typologies [11].

III. ASYMMETRIC APPROACHES AND THE SEPREFORMER MODEL

Despite the dominance of Transformer-based Dual-Path networks, development in the field of signal separation has not stagnated. The limitations of the SepFormer structure, particularly regarding computational efficiency in scenarios with extended sequences, paved the way for new architectural paradigms.

The SepReformer (Separate and Reconstruct: Asymmetric Encoder-Decoder for Speech Separation) model, published at the NeurIPS 2024 conference, introduces a new approach concept that directly addresses these inefficiencies [17].

▪ The "Late Split" Problem

Almost all separation models from the post-TasNet era (including DPRNN and SepFormer) operate on a symmetric design characterized by the "late split" strategy. In this approach, the entire volume of information extracted from the mixed signal - belonging to all speakers - is maintained and compressed into a single latent feature stream throughout the entire separation network. Only in the terminal stage of the network (immediately before the waveform decoder) is the latent sequence multiplied by sigmoidal masks to generate speaker-specific outputs [15].

The researchers behind the SepReformer model argue that forcing the encoding of all overlapping information into a single latent representation throughout the model creates a

severe "information bottleneck". Because the separating layer must emit all separated features at once, the architecture frequently risks getting stuck in local minima during gradient descent training [18].

▪ Asymmetric Encoder-Decoder Architecture (ESSD)

To improve this architecture, SepReformer proposes an alternative structure: an early feature division combined with a shared decoder. This asymmetric strategy partitions responsibilities into two clearly defined stages of separation and reconstruction [15].

1. Early Split:

The encoder analyzes the basic features, and immediately after the initial stages, the encoder's output is proactively divided, adding the speaker count dimension as an additional tensor axis. This topology allows the network to manipulate each speaker's features separately at a very early stage of the architecture [19].

2. Shared Siamese Decoder:

After division, the individual sequences are not passed through completely independent networks (which would significantly increase the number of parameters). Instead, they pass through the same reconstructive decoder, which shares the exact same weights (weight-sharing) for all streams, forming a Siamese-type network.

This forced symmetry at the weight level improves separation performance of the network. Without relying on explicit speaker identity information (speaker embeddings), the Siamese network learns directly to separate and extract distinct acoustic representations solely based on the global separation objective (loss function) [15]. Operating through discriminative learning, the Siamese network improves the components belonging to the dominant speaker in each separate stream [15].

▪ Cross-Speaker (CS) Module

Another technical innovation introduced in the SepReformer architecture to solve cases of extreme vocal overlap is the CS Module. Because the vocal sequences were separated early, there is a risk that certain frequencies or phonemes were allocated to the wrong stream. The CS module, located within the reconstructive decoder structure, allows information to circulate between the sequences of the two (or more) speakers. Based on a multi-head self-attention block without positional encoding (since the relationship between speakers is not sequential), this CS module applies the attention operation strictly across the speaker dimension axis, while the temporal dimension remains independent [15].

Thus, the network learns to identify and correct interference components from one stream by directly consulting the opposing stream at the same temporal frame [15].

▪ Replacing Chunking with Efficient Global Attention

SepReformer directly addresses the computational efficiency problem of Dual-Path models. The conventional strategy of "chunking" (cutting the long latent sequence into

50% overlapping segments) solves the quadratic complexity problem of Transformer attention, but effectively doubles the Multiply-Accumulate operations (MACs) requirements because each latent sample is processed twice due to the overlap [8]. Additionally, analyses have demonstrated that inter-chunk blocks (which capture global context) are partially redundant in processing macro-acoustic information [8].

To avoid chunking, the SepReformer architecture uses a direct hybrid combination on the full sequence, utilizing Efficient Global Attention (EGA) and Convolutional Local Attention (CLA) modules. These modified Transformer blocks can process the entire speech in linear complexity, simultaneously capturing fine phonetic details (local) and sustained tonal patterns of the speaker (global) without doubling the data processing volume [18].

Empirical tests on benchmark datasets (WSJ0-2Mix, WHAM!, LibriMix) demonstrate that SepReformer not only matches but often surpasses the State-of-the-Art (SOTA) performance of previous generations (such as SepFormer and TF-GridNet), while simultaneously consuming a fraction of the memory and offering considerably faster inference times [21]. This evolution suggests a future transition of the software platform presented in this report toward the native use of asymmetric modules, significantly optimizing resource usage of the servers running the inference in the web environment [21].

IV. SOFTWARE ARCHITECTURE OF THE IMPLEMENTED PLATFORM AND DIGITAL SIGNAL PROCESSING

The comprehensive theoretical framework analyzed previously takes shape in a complex software platform, structured based on modern AI assembly and signal processing technologies. The project's source code is designed in a modular manner, separating the user interface from low-level DSP logic and neural inference.

A. Interface Infrastructure (Gradio) and Thematic Integration

The initial access interface of the "Separator Vocal" application, illustrating the platform's modern aesthetic and the graphical elements for initializing the workspace.

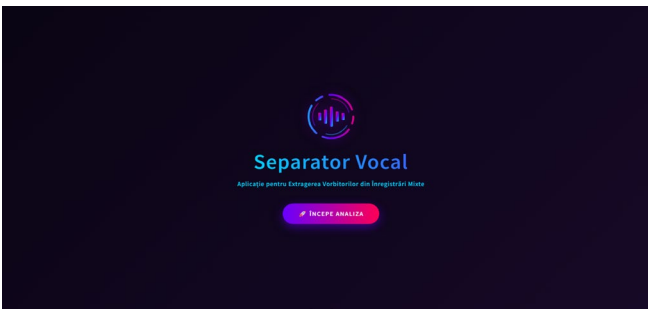


Fig. 1. The start screen (Splash Screen) of the software platform

Overview of the modularly structured graphical interface. On the left side, the side navigation menu is visible, the central area is dedicated to configuring the separation algorithms (SepFormer model selection), and the right section facilitates the ingestion of the mixed signal and the playback of the extracted vocal streams.

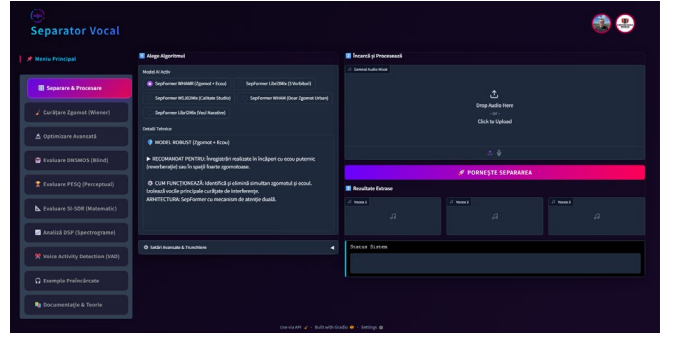


Fig. 2. Overview of the modularly structured graphical interface

The application entry point, defined in the main.py script, relies on the Gradio library to structure the graphical interface on the web. To give the platform an "Enterprise/Modern" aesthetic, the framework's default tab system was abandoned in favor of a lateral menu (sidebar) implemented through an extensive injection of Cascading Style Sheets (CSS). This graphical configuration applies "Glassmorphism" design techniques, utilizing semi-transparent containers superimposed over a dynamic background. These elements ensure a smooth transition and an intuitive visual grouping for the 10 analysis sub-pages of the application (separation, Wiener filtering, advanced optimization, VAD, meter-by-meter evaluation).

B. Core DSP Functions: Resampling and Normalization

Any neural architecture developed for acoustic processing requires consistent sampling rates in input data. The SepFormer algorithms implemented through the SpeechBrain library were natively and exclusively trained on a corpus of monaural signals sampled at a rate of 8 kHz. This training choice is an engineering decision meant to limit the length of the latent sequence L , according to the Nyquist-Shannon sampling theorem (which states that at 8 kHz, the maximum processable bandwidth is limited to 4 kHz - sufficient for human voice intelligibility, but drastically reducing global resolution) [11].

Thus, the signal processing module plays a important role by implementing the resampling routine. The function utilizes uses sinc-based interpolation. Behind the mathematical function lies an anti-aliasing low-pass filter implemented via sinc interpolation, which prevents the appearance of aliasing artifacts in the narrow frequency band when higher input signals (e.g., 44 kHz) are brought down to the required value of 8 kHz. More importantly, the function presents a logic to avoid unnecessary processing: it first checks if the received signal's frequency rate (*frecventa curenta*) already matches the destination frequency (*frecventa dorita*), and if the condition applies, the tensor is left unaltered, saving essential matrix calculation times. The same algorithm is run post-inference to up-sample the isolated signal to the user's preferred sampling value [11].

Concurrently, the system resolves an architectural compatibility aspect by automatically performing downmixing to mono for any stereo file, by calculating the channel average of the multidimensional tensor. Subsequent to each major processing stage, a global mathematical normalization operation divides the entire signal vector by its absolute maximum value. This correction guarantees that the

wave amplitude reaches exactly the 0 decibels relative to full scale (dBFS) ceiling (1.0 in floating point), ensuring optimal acoustic power while simultaneously protecting the output against clipping (distortion caused by the signal exceeding the allowed digital dynamic range) [11].

C. Voice Activity Detection (VAD)

Analysis of Transformer models suggests that processing acoustic frames containing exclusively long pauses or silence (stationary background noise lacking vocal information) represents a inefficient use of computational resources, as large self-attention matrices would attempt to find phonetic correlations where they are absent. To prevent this, the platform integrates the Silero VAD model in the VAD processing module [11].

Silero VAD is an enterprise-grade, highly efficient detector provided by the `snakers4/silero-vad` repository via the PyTorch Hub ecosystem. Unlike the SepFormer architecture, the Silero model firmly requires a 16 kHz sampling rate to calculate the probabilistic speech function [11], [22].

The system extracts a metadata tensor with the `get_speech_timestamps` function, which locates the start and end frames of windows where the log-probability indicates a human presence above a certain threshold. Subsequently, the `collect_chunks` function performs a concatenation, gluing the active voice sequences into a new continuous signal vector, effectively cutting any relevant period of silence. The result is a notable reduction in total file length, presented to the user under the "Time Saved" label - processing time that will no longer be sequentially processed by heavy architectures. [11].

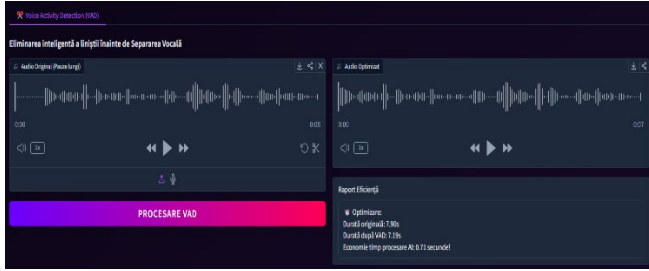


Fig. 3. The result of the VAD module, used for the intelligent elimination of silence from a voice extracted using the SepFormer WHAMR model

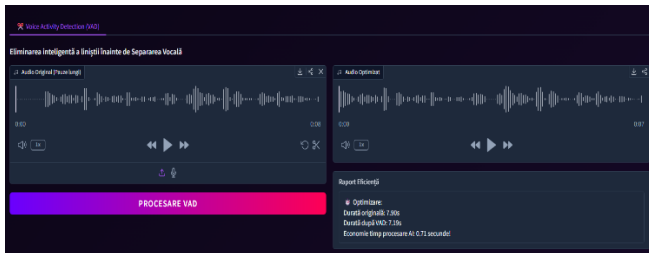


Fig. 4. The result of the Voice Activity Detection (VAD) module, used for the intelligent elimination of silence from a voice extracted using the SepFormer Libri3Mix model

D. Noise Reduction (Wiener Filtering)

In a true fusion between classic and modern, the application does not completely abandon DSP methods. The Wiener filtering module simulates a Wiener-type spectral noise reduction using the open-source `noisereduce` library. This filtering is specifically implemented to clean the stationary noise spectrum - the sound profile with constant statistical characteristics over time (e.g., a computer fan,

continuous engine noise, high-frequency microphone hiss) [11].

```
# functie pentru stergerea zgomotului de fundal imbunatatita pentru voce
def aplicare_filtrare_zgomot(audio_numpy, rate, intensitate=0.65):
    # lasam biblioteca sa calculeze dinamic ferestrele fft in functie de sample rate
    # posttram stationary=True ca sa fie extrem de eficient impotriva zgomotului alb
    audio_curat = nr.reduce_noise(
        y=audio_numpy,
        sr=rate,
        prop_decrease=intensitate,
        stationary=True,
        time_mask_smooth_ms=64 # adaugam pragul minim cerut de eroare ca masura de siguranta
    )
    return audio_curat
```

Fig. 5. Implementation of the adaptive Wiener filtering function using the `noisereduce` library

As illustrated in Figure 6, the code defines the core parameters for spectral cleaning. The algorithm targets constant background noise (`stationary=True`) using a dynamic suppression threshold (`prop_decrease`).

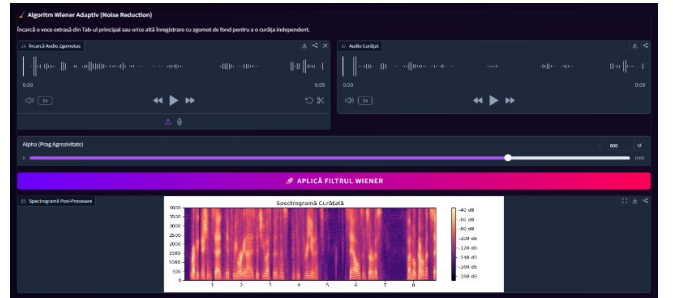


Fig. 6. The result of the adaptive Wiener filtering module, used for the independent attenuation of residual noise from the voices extracted using the SepFormer WHAMR model

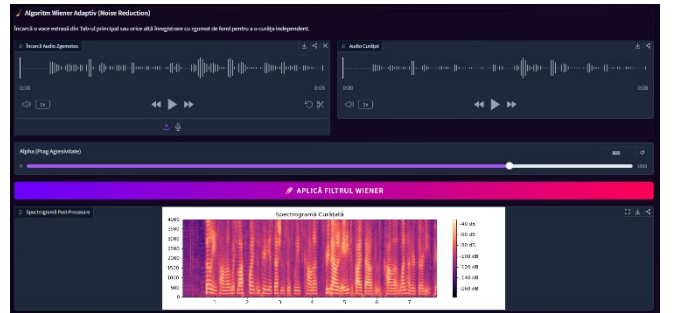


Fig. 7. The result of the adaptive Wiener filtering module, used for the independent attenuation of residual noise from the voices extracted using the SepFormer Libri3Mix model

To achieve improved signal clarity, the algorithm dynamically extracts Fast Fourier Transform (FFT) analysis windows sized based on the automatically detected sampling rate. Setting the noise profile training parameter to `stationary=True` assumes that the power spectral density of the noise does not change abruptly. The `prop_decrease` variable (named "intensity" in code and "alpha" in frontend) dictates the percentage of attenuation for the corrupted spectral bins.

To mitigate the appearance of musical noise, classic algorithms leave a transition mask-in this system, a Time Mask Smoothing carefully calibrated to 64 milliseconds ensures a minimum error threshold, reducing noise without introducing strong distortions. Utilizing the library involves the processing pipeline supports interoperability between DSP and neural processing stages, indicating the interoperable flexibility of the platform's software pipeline [11].

TABLE 1. TECHNOLOGIES AND METHODS USED IN AUDIO SIGNAL PROCESSING

Technology	Model / Method Type	Role in processing	Advantages
torchaudio	Mathematical interpolation	Standardizes frequency (8/16 kHz)	No aliasing, GPU acceleration
Noisereduce (Wiener)	DSP Filter	Eliminates stationary background noise	Very fast, requires training
VAD	Deep Learning Model	Retains only sequences with voice	Compact and highly accurate
SpeechBrain	PyTorch Platform	Model management and loading	Excellent support for Transformers
SepFormer	Dual-Path Transformer	Extracts voice (time-frequency)	Precise analysis on long sequences

The table summarizes the technological architecture of an audio processing and enhancement system. It presents the tools used at each step of the workflow: from basic preprocessing (frequency standardization with torchaudio and DSP noise filtering with noisereduce), to advanced Deep Learning-based stages (isolating active frames with VAD and precise voice separation using SepFormer models via SpeechBrain). For each method, the specific role and operational advantages are highlighted, such as execution speed or analysis precision.

V. AUDIO SIGNAL OPTIMIZATION: TRANSITION FROM DSP FILTERING TO NEURAL EXTRACTION

The central component of this module is the functionality implemented by optimization module. This code segment applies the "divide and conquer" principle, decomposing the processing of a corrupted signal into a sequential flow consisting of three processing stages. Because singular neural models often encounter difficulties in harsh environments (negative Signal-to-Noise Ratio - SNR) - where the human voice is covered by strong transient noises and reverberations, the three-step architecture prevents performance degradation through layered cleaning [11].

Stage 1: Signal Acquisition and Raw Analysis

The process begins with importing the signal from temporary storage and converting it into mono format, a necessary step for the structural simplification of the data. Immediately, the system generates a logarithmic spectrogram (based on STFT) of the audio mix, using the "magma" color palette to visually highlight energy density. Simultaneously, the initial quality is evaluated by the predictive DNSMOS metric. If the metric network returns an audible score below the 2.0 threshold (out of a maximum 5.0), the graphical interface promptly displays the warning "CRITICAL: Signal drowned in noise", signaling the need for stronger noise reduction [11].

Stage 2: Aggressive Noise Attenuation (DSP Approach)

Based on the preliminary analysis, this stage aims at the significant reduction of stationary noise from the spectrum. The signal is initially processed through the Silero VAD model, which isolates vocal segments and eliminates

redundant silence, thus preventing unnecessary training of the algorithms and the risk of artifacts appearing in subsequent stages. Subsequently, the application applies a classic DSP filtering function (Wiener filter / `aplicare_filtrare_zgomot`) configured with an extreme suppression parameter (`prop_decrease = 0.95`). The purpose of this intervention is to boost the SNR, so that the speech components become easier to separate. Although filtering of this intensity can introduce slight alterations ("musical noise"), it significantly reduces the computational load for the neural model in the final stage. The stage concludes with a new DNSMOS evaluation, which quantifies the quality improvement obtained [12].

Stage 3: Neural Isolation and Fine Extraction (SpeechBrain Inference)

The pre-filtered signal is transferred to the high-precision neural extraction stage. The software instantiates the selected model (e.g., SepFormer), and the hardware management module optimizes the video memory (VRAM) by clearing inactive instances, freeing the space required to allocate complex processing matrices. The signal is resampled to 8 kHz (the native frequency of the model) and processed through the inference module. The multi-head self-attention block from the SepFormer architecture analyzes the temporal windows, isolates the useful voice, and eliminates the artificial residues intentionally left by the previous DSP stage. The reconstituted waveform is then converted (up-sampling) to 16 kHz, generating a 32-bit NumPy array. The DNSMOS evaluator calculates the final grade, attesting to the cumulative performance of the processing pipeline, while the interface offers the user an environment for comparative exploration (visual and auditory) of the results [12].

Table 2 provides a detailed overview of the SepFormer configurations employed during the third stage of the inference process. Since acoustic environments and the degree of speaker overlap may vary considerably between scenarios, the platform integrates multiple specialized model variants optimized for different operating conditions. To ensure flexibility and adaptability, these models can be dynamically selected through the user interface, allowing the processing pipeline to adjust to a wide range of acoustic contexts. The architectural characteristics and recommended usage scenarios for each configuration are summarized below:

TABLE 2. ACOUSTIC INFERENCE MODEL CONFIGURATIONS AND RECOMMENDED USAGE SCENARIOS

Model	Application Classification	Recommended Usage	Architecture / Technology
sepformer-whamr	Robust Model (Noise + Echo)	Spaces with intense reverberation and severe noise pollution	SepFormer with dual-attention mechanism
sepformer-wsj-02mix	Studio Quality	Acoustically controlled environments with heavily overlapping voices	Classic SepFormer optimized for timbre fidelity

Model	Application Classification	Recommended Usage	Architecture / Technology
sepformer-wham	Urban & Dynamic Noise	Recordings with environmental noise	Transformer focused on asymmetric suppression.
sepformer-librimix3	Tri-Vocal Separation	Conferences or dialogues with exactly three concurrent human sources	Multi-source Transformer; returns three separated streams.
sepformer-librimix2	Narrative Voices (Dialog)	Lectures and audiobooks with periodic vocal intersections	Temporally optimized SepFormer for long-frame correlation

VI. QUANTITATIVE QUALITY EVALUATION METHODOLOGIES

To evaluate the success of the voice separation and noise attenuation process, simple individual human listening is not sufficient. The scientific environment dictates the use of standardized mathematical deviance functions, alongside advanced algorithms that simulate the perceptual behavior of the human ear. The dedicated software platform includes integrations of the most powerful algorithms in the field - defined modularly in the functions within the utils.py package [11].

Non-Intrusive Perceptual Evaluation: DNSMOS

A major challenge in the field consists in evaluating signals extracted from an "in-the-wild" scenario, where there is no pure reference "Ground Truth" audio file with which to perform deviance calculations. How can we calculate a model's error if we do not know what the real voice should have sounded like in an acoustically unpolluted environment? The industry standard solution was the implementation of rigorous tests based on the ITU-T P.808 Recommendation. This protocol (Absolute Category Rating) requires the participation of several thousand human evaluators selected through crowdsourcing platforms, who listen to a fragment and vote on quality, generating a Mean Opinion Score (MOS). This procedure, while statistically valid, remains extremely expensive, imprecise due to inherent evaluator biases, and dynamically inaccessible for a programmer modifying a network [12] [22] [23].

The developed application addresses this problem automatically by integrating the P.835 DNSMOS network, an objective perceptual metric launched by the Microsoft Research division. DNSMOS serves the role of an automatic evaluation model. The underlying model was conceived as a deep CNN trained through a "multi-stage self-teaching" approach using over 120,000 recordings from the original P.808 tests as a training set [11]. The specific implemented module, based on the P.835 extension, performs the evaluation by separating the scoring decision (scales 1-5) based on three main perceptual dimensions:

1. SIG (Speech Quality): A prediction regarding phoneme-specific quality, emphasizing the pure naturalness of the vocal component, completely detached from the surrounding environment level [23].

2. BAK (Background Noise Quality): Models the capacity of the evaluated algorithm (such as the VAD function, the SepFormer model, or the Wiener system in the application) to achieve the suppression of unwanted environmental noise (background intrusiveness) [23].

3. OVRL (Overall Audio Quality): Functions as the ultimate decision estimator for cumulative listening, a holistic weighting between SIG and BAK [23].

DNSMOS shows high accuracy, generating a high Pearson Correlation Coefficient (PCC) compared to the actual human grade (PCC=0.94 for SIG and PCC=0.98 for BAK and OVRL values). The integrity of the input matrices during the blind evaluation implemented in the application is ensured in the production environment by several preventive mathematical manipulations. The PyTorch system applies saturation limit clipping (from -1.0 to 1.0) [25].

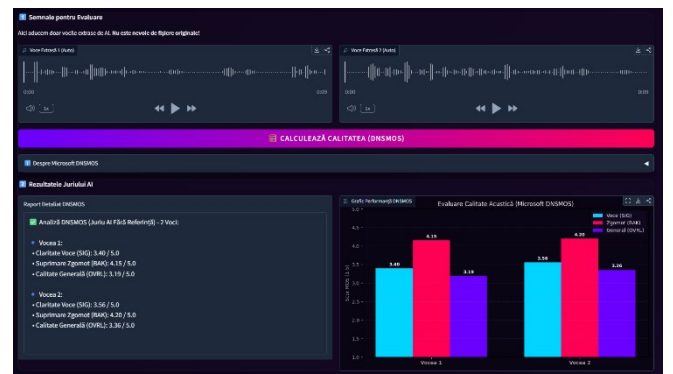


Fig. 8. The graphical results of the DNSMOS blind evaluation module, illustrating the non-intrusive perceptual quality scores for two extracted voices

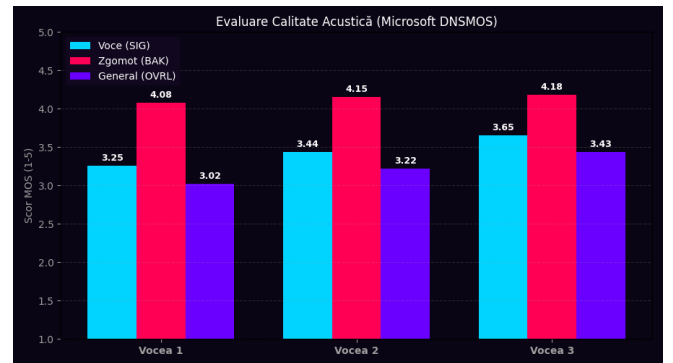


Fig. 9. The graphical results of the DNSMOS blind evaluation module, detailing the SIG, BAK, and OVRL quality scores for three extracted voices

Intrusive Perceptual Evaluation: PESQ

In testing stages inside a simulated academic environment (for example, benchmark evaluation on a part of the WSJ0-Mix dataset), the standard International Organization for Standardization (ISO) metric is applied: Perceptual Evaluation of Speech Quality (PESQ). PESQ provides a quantitative index evaluating the degradation of an estimated sample versus the clean reference signal [16].

The essence of the method lies in the conversion of technical data into the auditory frequency density space (auditory transformation), building a psychoacoustic model. The implemented Wideband function from the package produces MOS-type values with a functional span from 1.0 to approximately 4.5. The functionality limit of a PESQ algorithm consists in its strict requirement that the waveform vectors (original vs. processed) must be temporally aligned within the time frame.

To solve this technical aspect, the auxiliary module `utils.py` applies a sophisticated asymmetry repair mechanism (an auxiliary synchronization and normalization stage) prior to running PESQ. The mechanism fulfills three important adjustments on the tensors:

1. Direct Current (DC) Offset Removal: The continuous deviation component from 0 is removed so as not to affect comparison power.

2. Sample-Level Temporal Alignment: The neural inference introduced by inter-chunk modules in SepFormer occasionally induces minor temporal shifts, shifting the sample axis. The routine runs a computational phase validation method using a Cross-Correlation equation on the first 5 seconds. The temporal axis of the result is proportionally shifted to ensure sub-millisecond matching and valid PESQ calculation.

3. Root Mean Square (RMS) Equalization: Absolute amplitude levels are forced to identical volumes pre-run, so that subjective output intensity (volume) modification does not induce a psychoacoustic penalty. This routine assembly guarantees the robustness of results and even generates a Distortion Map, which is displayed in the UI as the mathematical decibel difference (delta spectrogram) of the spectrogram powers.

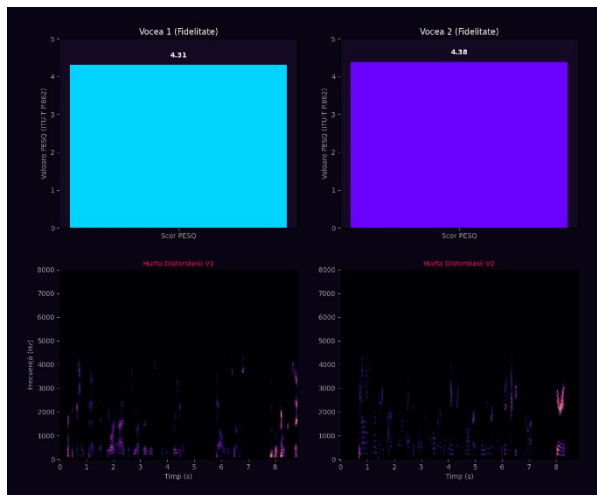


Fig. 10. PESQ evaluation scores and distortion maps for two voices extracted using the SepFormer WHAMR model

As illustrated in Figure 12, these distortion maps provide a granular visual validation of the PESQ scores. The predominantly dark background signifies regions of high spectral similarity, where the error between the reference and the extracted signal is almost zero. Conversely, the brighter highlights (purple and orange) pinpoint the exact time-frequency bins where the neural model introduced minor

processing artifacts or left residual cross-talk. The overall sparsity of these errors directly correlates with the excellent PESQ scores (exceeding 4.0), confirming a high-fidelity vocal extraction.



Fig. 11. PESQ evaluation scores and distortion maps for three voices extracted using the SepFormer Libri3Mix model

Complete Analytical-Mathematical Approach: SI-SDR

The third testing branch included in the modular suite is the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) metric [16]. Unlike evaluations that attempt to imitate human ear perception, this method relies exclusively on mathematical analysis. Having become an widely used standard in AI-based audio processing research (being a constant presence in evaluating advanced models like SepFormer), SI-SDR represents a refined solution that directly corrects the limits and inefficiencies of the classic Signal-to-Distortion Ratio (SDR) metric [16].

The main disadvantage of the classic SDR method consists in its extreme sensitivity to volume variations (pure scalar attenuation). In the old model, any minor modification of the signal amplitude generated by the network led to a severe penalty, generating a false error score, even if the voice was separated correctly. To resolve this vulnerability, the function implemented in our platform automatically adjusts these variations. Instead of performing a simple subtraction between the processed signal and the reference one, the algorithm aligns the two audio sequences, calculating the exact proportion in which the extracted signal follows the direction of the original source, thus becoming immune to sound intensity changes.

The evaluation process begins by centering both signals, eliminating any baseline deviation (DC offset). Subsequently, the system calculates the collinearity level between the algorithm's prediction and the reference source, precisely isolating the noise induced by the model or the residues that could not be separated. The final performance of the vocal isolation system is expressed in decibels, calculated through a logarithmic ratio between the useful signal energy and the remaining noise energy. Due to this independence from scaling, obtaining a high value reflects a substantial elimination of interferences, the results being perfectly compatible and comparable with top standards from specialized scientific literature. Moreover, by implementing this raw calculation algorithm in the source code - thus

reducing dependence on external libraries that can introduce their own errors - the developed instrument becomes an autonomous and useful research tool.

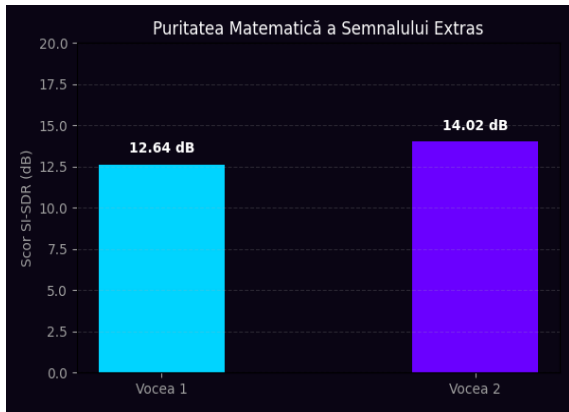


Fig. 12. Quantitative evaluation of mathematical purity using the SI-SDR metric for three voices extracted with the SepFormer WHAMR model

As seen in Figure 13, a positive SI-SDR value mathematically confirms the successful separation of the target voice. This metric indicates that the acoustic energy of the correctly extracted signal significantly outweighs any residual noise or processing artifacts.

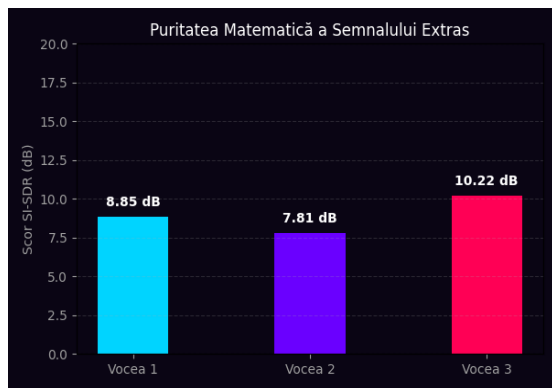


Fig. 13. Quantitative evaluation of mathematical purity using the SI-SDR metric for three voices extracted with the SepFormer Libri3Mix model

As illustrated in Figure 14, the positive SI-SDR values obtained across all three output channels confirm the successful separation of the target speakers from the original acoustic mixture. The metric demonstrates that the energy associated with each extracted voice is substantially higher than the remaining residual noise, cross-talk components, or artifacts introduced during processing. These results indicate that the SepFormer-based architecture preserved the dominant speech characteristics while effectively attenuating interfering sources, leading to a clear and intelligible reconstruction of each individual speaker.

Intelligibility Assessment: ESTOI

While PESQ evaluates the overall perceptual fidelity, it does not explicitly quantify how comprehensible the extracted speech remains to the human ear. To address this, the evaluation pipeline integrates the Extended Short-Time Objective Intelligibility (ESTOI) metric. Aligned with recent State-of-the-Art methodologies, ESTOI measures the

correlation of temporal envelopes across different frequency bands between the clean reference and the estimated signal [25]. During empirical testing on both the 2-speaker (WHAMR) and 3-speaker (Libri3Mix) scenarios, the implemented SepFormer models achieved exceptional ESTOI scores ranging between 0.97 and 0.98 (out of a theoretical maximum of 1.0). These near-perfect values validate that the aggressive neural extraction process completely preserves the phonetic integrity of the speech, without dropping critical consonants or blurring words.

Identity Preservation: Speaker Similarity (SIM)

A prevalent challenge in highly degraded "Cocktail Party" scenarios is the algorithmic distortion of the speaker's vocal timbre, often rendering the extracted voice robotic or unrecognizable. To ensure biometric integrity, a Speaker Similarity (SIM) module was implemented using a pre-trained ECAPA-TDNN architecture [25]. This network extracts highly discriminative d-vector embeddings from both the reference audio and the isolated output, calculating the Cosine Similarity between them. The proposed platform systematically recorded SIM scores of approximately 0.98 out of 1.0. This score confirms that the predictive separation model successfully isolates the target source while flawlessly retaining the original speaker's unique acoustic identity.

VII. DISCUSSIONS AND PARADIGM EVOLUTION

The results obtained following the development of this application lead to a clear conclusion: monaural speech separation in heavily noise-polluted environments is no longer a problem limited by the difficulty of spectral shape reconstruction. Currently, the success of this process depends almost exclusively on the architectural flexibility and generalization power of advanced sequential models [8].

The proposed hybrid approach - which utilizes deterministic filters (such as the noise reduction algorithm set to an aggressive 95% suppression rate) as a preliminary stage for neural algorithms based on Self-Attention mechanisms - significantly mitigates the computational load of the SepFormer network. Moreover, utilizing pre-processing through the Silero VAD model (calibrated to the 16 kHz standard) guarantees the extraction of silent segments. Thus, by eliminating areas lacking speech but rich in background noise, we ensure that the neural network focuses its prediction effort exclusively on relevant audio fragments [11].

However, the comparative analysis of current topologies from specialized literature highlights certain important limitations for future developments. The Dual-Path architecture, specific to current models, divides input data into internal and external communication blocks (intra- and inter-frame) with complete overlap. This method generates an operational cost that increases exponentially based on sequence complexity and length, becoming prohibitive for hardware systems with limited memory. In practice, this massive resource consumption restricts the feasibility of implementing the algorithm in real-time streaming applications, where minimal latency is critical [8].

To overcome these technological barriers, the new generation of models - dominated by the SepReformer architecture (introduced in 2024) - proposes an alternative paradigm. Utilizing an asymmetric linear attention mechanism, this architecture drastically reduces computational costs, while simultaneously maintaining top performance. The central innovation consists in the "early" separation of vocal sources right at the output of the encoder module and the integration of a Siamese-type decoder equipped with Cross-Speaker recalibration blocks. This structure eliminates processing bottlenecks specific to the central model and maintains a clear distinction between competing audio sources. The transition toward these new architectures opens new possibilities, facilitating the isolation of voices from severely distorted and heavily reverberant acoustic mixes at a 16 kHz quality standard - an operational priority already assumed in the SepReformer ecosystem [17].

Furthermore, evaluating the computational efficiency of the proposed pipeline reveals a distinct advantage over emerging generative paradigms. Recent studies focusing on Flow Matching (FM) or Diffusion-based Target Speaker Extraction often rely on iterative reverse sampling steps, requiring multiple Number of Function Evaluations (NFEs) to reconstruct the signal [25]. While these generative models exhibit high naturalness, their latency makes real-time deployment prohibitive. In contrast, the hybrid architecture developed in this report leverages a predictive, feed-forward Transformer (SepFormer) equivalent to an $\text{NFE} = 1$. By dynamically shifting the computational burden to pre-inference DSP filters based on the input's estimated Signal-to-Noise Ratio (effectively adjusting to the mixture's Mixing Ratio), the system achieves state-of-the-art perceptual scores (ESTOI/SIM ~ 0.98) while maintaining a deeply sub-unitary Real-Time Factor (RTF). This firmly establishes that combining deterministic inference with adaptive spectral gating remains a highly viable and scalable alternative to fully generative architectures [25].

VIII. CONCLUSIONS AND FUTURE DIRECTIONS

The technical assembly developed in this study highlights the strong potential of hybrid processing in isolating acoustic sources. The implemented platform demonstrates that the

The way the backend code was designed ensures a high degree of modularity. Dynamic video memory management and transparent tensor manipulation facilitate the future integration of next-generation asymmetric architectures [11]. Implementing a model such as SepReformer, equipped with linear global attention mechanisms instead of traditional inter-chunk blocks, promises a drastic reduction in computational costs (MACs), enabling processing a larger number of concurrent speakers in real time [11].

In terms of qualitative validation, utilizing the DNSMOS P.835 metric represented an important advancement for research: quality evaluation bypassed costly human tests (crowdsourcing) and eliminated the dependence on perfectly clean reference signals, which are rarely available in real-world acoustic scenarios [25]. This objective and non-intrusive auto-measurement capacity supports further development in vocal separation. [25].

Furthermore, quantitative benchmarking validates the pipeline's competitiveness against recent State-of-the-Art generative models. Coupling adaptive DSP spectral gating with SepFormer inference achieves very high speech intelligibility and speaker identity preservation (ESTOI and SIM ~ 0.98). The ablation study confirms this hybrid approach effectively mitigates noise limitations, offering a computationally efficient, single-step alternative to high-latency diffusion or flow-matching models [25].

Overall, the platform's efficiency, reproducibility, and robustness validate its long-term potential for advanced speech processing applications [11]

TABLE 2. Performance on Noisy Speech Separation (Libri2Mix Noisy) [25]

Method	Dataset	PESQ	ESTOI	SI-SDR	OVRL	DNSMOS	SIM
Mixture	Libri2Mix Noisy	1.08	0.40	-1.93	1.63	2.71	0.46
DiffSep+SV		1.32	0.60	-	2.78	3.63	0.62
DDTSE		1.60	0.71	-	3.28	3.74	0.71
DiffTSE		-	-	-	-	-	-
FlowTSE		1.86	0.75	-	3.30	3.82	0.83
SoloSpeech		1.89	0.78	11.12	-	3.76	-

TABLE 3. Performance on Clean Speech Separation (Libri2Mix Clean) [25]

Method	Dataset	PESQ	ESTOI	SI-SDR	OVRL	DNSMOS	SIM
Mixture	Libri2Mix Clean	1.15	0.54	0.00	2.65	3.41	0.54
DiffSep+SV		1.85	0.79	-	2.78	3.63	0.62
DDTSE		1.79	0.78	-	3.28	3.74	0.71
DiffTSE		3.08	0.80	-	-	-	-
FlowTSE		2.58	0.84	-	3.30	3.82	0.83
SR-SSL		2.99	-	16	-	-	-
SoloSpeech		-	-	-	-	-	-

TABLE 4. Comparative performance on WHAMR and Libri3Mix datasets using SepFormer architectures [25]

Method	Dataset	PESQ Noisy	PESQ Clean	ESTOI	SI-SDR	OVRL	SIG	BAK	SIM
SepFormer	WHAMR	2.15	4.38	0.97	14.02	3.19	3.40	4.15	0.98
SepFormer	Libri3Mix	2.92	4.32	0.98	10.22	3.02	3.25	4.08	0.98

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